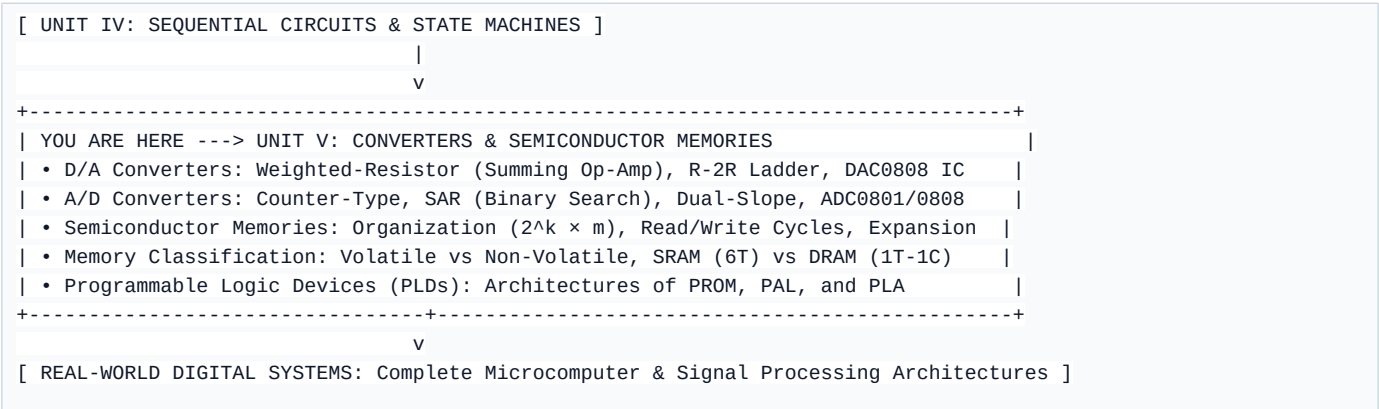


DEC Unit V: Converters & Semiconductor Memories

Comprehensive First-Principles Lecture Notes

Unit V: Converters and Semiconductor Memories

Master Subject Map: Unit V Position



In 3 lines: Units I through IV built the internal brain of a digital computer. Unit V completes the subject by connecting that digital brain to the physical universe. We master **DACs and ADCs** to translate between continuous real-world voltages and digital bits, explore mass **Semiconductor Memories (RAM & ROM)**, and examine user-customizable silicon chips (**PLDs: PROM, PAL, PLA**).

Prerequisites for this Unit

Before studying Unit V, ensure you recall these concepts: 1. **Ohm's Law & Circuit References (Prereqs 1 & 2):** Current $I = V/R$, Ground (0 V), and Op-Amp Virtual Ground ($V_- \approx 0\text{ V}$). 2. **Binary Weights & Positional Values (Unit I, Section 1.1):** Binary digits carry weights $2^{\{-1\}} = 0.5$, $2^{\{-2\}} = 0.25$, $2^{\{-3\}} = 0.125$, etc. 3. **Binary Counters & Shift Registers (Unit IV, Sections 4.8–4.10):** Stepping sequentially through states on clock edges. 4. **Decoders (Unit III, Section 3.8):** A k -to- 2^k decoder activates exactly one row wire for memory word addressing.

PART 1: Digital-to-Analog Converters (DAC)

5.1 The Need for DAC and Key Specifications

Source: Kumar (Ch. 17, pp. 941–946, Figures 17.1 & 17.2); Floyd (Ch. 9, pp. 487–492)

[Pacing: Intuitive Foundations — Bridging Bits to Continuous Reality]

Need (Problem-First)

Microprocessors calculate with discrete numbers (e.g., 11010010). But physical actuators do not understand numbers. An audio speaker requires a continuously undulating analog voltage to move its cone; an electric vehicle motor requires a continuous voltage to set its speed; an MRI scanner requires continuous magnetic coils. How can a computer convert a digital binary word into a proportional continuous physical voltage? That bridge is the **Digital-to-Analog Converter (DAC)**.

Key DAC Performance Specifications Defined Inline:

1. **Resolution:** The smallest analog voltage increment that the DAC can produce. It corresponds to the voltage generated by a change of the **Least Significant Bit (LSB)** (Kumar, p. 944):

$$\text{Step Size } (\Delta V) = \frac{V_{\text{FS}}}{2^n - 1}$$

Where V_{FS} is the Full-Scale output voltage and n is the number of digital bits. *Concrete Round-Number Example:* An 8-bit DAC ($n=8$) has $2^8 - 1 = 255$ intervals. If $V_{\text{FS}} = 5.10 \text{ V}$, the resolution is:

$$\Delta V = \frac{5.10 \text{ V}}{255} = 0.02 \text{ V} = 20 \text{ mV per step}$$

1. **Full-Scale Output (V_{FS}):** The maximum analog output voltage produced when all input bits are **1** (1111...1).
2. **Linearity:** How strictly the analog output follows a perfect straight line across all digital input codes.
3. **Accuracy:** The deviation between the actual measured analog output and the ideal theoretical output.
4. **Settling Time (t_s):** The time required for the analog output to settle within $\pm (1 / 2) \text{ LSB}$ of its final value after an input code transition.

5.2 The Binary Weighted-Resistor DAC

Source: Kumar (Ch. 17, pp. 951–953, Figure 17.10)

[Pacing: Derivation Step-by-Step — Trace the Summing Node]

How It Works (The Physical Mechanism)

The circuit connects an Operational Amplifier (Op-Amp) configured as an **inverting summing amplifier**, with binary input switches driving resistors whose values are scaled inversely to their binary weights (**R**, **2R**, **4R**, **8R**):

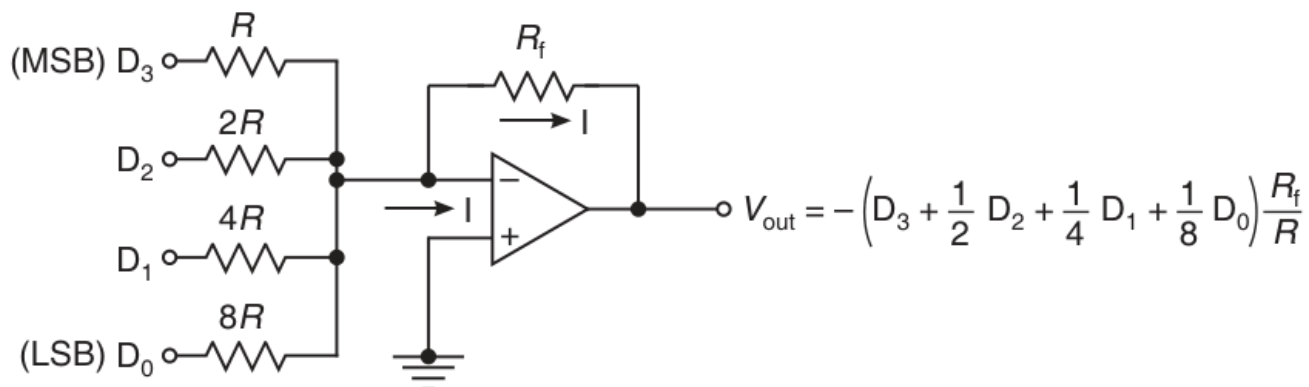


Figure 17.10 Weighted-resistor type DAC.

Figure

17.10: Binary Weighted-Resistor type Digital-to-Analog Converter using an operational amplifier summing network. Sourced from A. Anand Kumar, Fundamentals of Digital Circuits (4th Ed.), p. 951 (printed p. 919).

Characteristic 13 Walkthrough: Hardware Reality Check

1. The non-inverting terminal (+) of the op-amp is connected to Ground (**0 V**).
2. Due to negative feedback through **R_f**, the inverting terminal (-) is held at **Virtual Ground (0.00 V)** (Kumar, p. 951).
3. Digital bits (**D₃**, **D₂**, **D₁**, **D₀**) control electronic switches. If bit **D_i = 1**, the resistor is connected to reference voltage **-V_{ref}**; if **D_i = 0**, the resistor is connected to ground (**0 V**).
4. The current flowing through each weighted resistor is determined strictly by Ohm's Law into virtual ground:
5. **I₃ = V_{ref} / R** **D₃** (MSB current)
6. **I₂ = V_{ref} / (2R)** **D₂ = (1 / 2) I₃**
7. **I₁ = V_{ref} / (4R)** **D₁ = (1 / 4) I₃**
8. **I₀ = V_{ref} / (8R)** **D₀ = (1 / 8) I₃** (LSB current)
9. Because the op-amp input draws zero current, by Kirchhoff's Current Law, all individual branch currents sum into the feedback resistor **R_f**:

$$I_{\text{total}} = I_3 + I_2 + I_1 + I_0 = \frac{V_{\text{ref}}}{R} \leq \frac{V_{\text{ref}}}{R} (D_3 + (1/2)D_2 + (1/4)D_1 + (1/8)D_0)$$

10. The output voltage is the drop across R_f :

$$V_{\text{out}} = -I_{\text{total}} \cdot R_f = -V_{\text{ref}} \left(\frac{R_f}{R} \right) (D_3 + (1/2)D_2 + (1/4)D_1 + (1/8)D_0)$$

The Fatal Engineering Flaw of Weighted Resistors: The Resistor Spread Problem

In a 4-bit DAC, resistors range from R to $8R$ (a factor of 8). But in a modern **12-bit DAC**, the resistors must range from R to $2^{11}R = 2048R$! - If $R = 10 \text{ k } \Omega$, the LSB resistor must be **20.48 M Ω** ! - **Fabrication Disaster**: In integrated circuit silicon manufacturing, it is virtually impossible to fabricate resistors spanning four orders of magnitude ($10 \text{ k } \Omega$ to $20 \text{ M } \Omega$) that track each other accurately across temperature changes. A tiny **0.05%** error in the MSB resistor completely swamps the entire LSB current!

5.3 The R-2R Ladder DAC: The Universal Solution

Source: Kumar (Ch. 17, pp. 946–950, Figure 17.5); Floyd (Ch. 9, pp. 490–492)

[Pacing: Conceptually Brilliant — Master Why Two Resistor Values Suffice]

Need (Problem-First)

We need a digital-to-analog converter that works for any number of bits (8, 12, 16, or 24 bits) but uses **only TWO precision resistor values** that can be effortlessly matched on a silicon die. That circuit is the **R-2R Ladder**.

The R-2R Ladder Circuit Architecture (Kumar, p. 947, Figure 17.5)

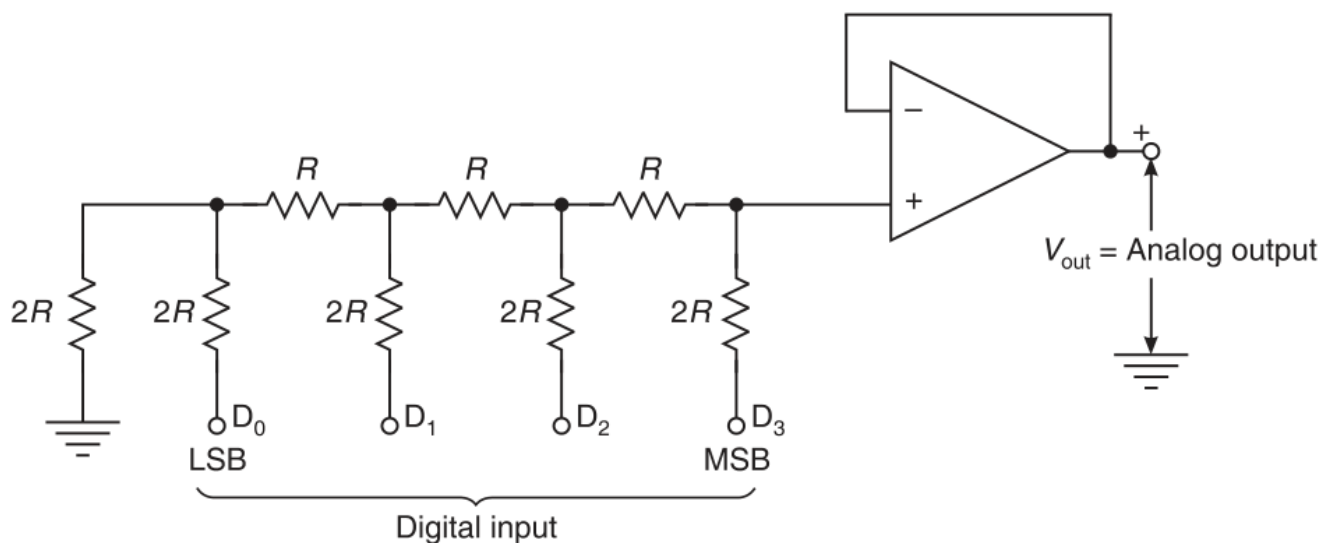


Figure 17.5 R-2R ladder type DAC.

Figure

17.5: Schematic diagram of an R-2R ladder Digital-to-Analog Converter feeding an operational amplifier buffer. Sourced from A. Anand Kumar, *Fundamentals of Digital Circuits (4th Ed.)*, p. 947 (printed p. 915).

Why It Works: The Continuous Binary Current Division

The R-2R ladder uses only two resistor values: R (e.g., $10 \text{ k } \Omega$) and $2R$ (e.g., $20 \text{ k } \Omega$).

- Look at the termination on the extreme left: a vertical $2R$ resistor is in parallel with another $2R$ resistor. Their equivalent parallel resistance is:

$$2R \parallel 2R = R$$

- This equivalent R is in series with the horizontal resistor R , yielding $R + R = 2R$!
- At the next node, this $2R$ is in parallel with the vertical $2R$, yielding R again!

- **The Binary Splitting Principle:** Looking into *any* node of the ladder toward the left, the equivalent resistance is **always exactly R!**
- Therefore, at every single node, incoming current splits **exactly in half (50% / 50%)!**
- The currents arriving at the summing bus are naturally, automatically weighted in exact powers of two:

$$(I / 2), (I / 4), (I / 8), (I / 16) \text{ dots}$$

- **Output Equation:**

$$V_{\text{out}} = V_{\text{ref}} \leq \text{ft}((D_3 / 2^1) + (D_2 / 2^2) + (D_1 / 2^3) + (D_0 / 2^4)\text{right})$$

- **Manufacturing Triumph:** Because only two resistor values are required, laser-trimmed thin-film silicon resistors maintain near-perfect tracking over temperature, making R-2R the foundation of commercial DACs.

Commercial DAC Example: DAC0808 / DAC0800 IC (Kumar, pp. 954–955)

An 8-bit monolithic multiplying DAC featuring an R-2R ladder, fast settling time (**150 ns**), and full 8-bit accuracy ($\pm 0.19\%$).

PART 2: Analog-to-Digital Converters (ADC)

5.4 The Need for ADC and Key Parameters

Source: Kumar (Ch. 17, pp. 957–968); Floyd (Ch. 9, pp. 492–502)

[Pacing: Intuitive Foundations — Translating the Analog World into Bits]

Need (Problem-First)

The real physical world is entirely analog. Sound waves entering a microphone, engine temperatures measured by a thermocouple, and blood pressure in a medical monitor are continuous, fluid voltages. A digital computer cannot directly process an infinite continuum of values. We must sample the continuous signal and quantize it into a sequence of binary words. That device is the **Analog-to-Digital Converter (ADC)**.

Key ADC Specifications Defined Inline:

1. **Resolution:** The number of bits in the output binary word (e.g., 8-bit, 12-bit, 16-bit). An **n-bit** ADC divides the analog voltage range into 2^n discrete quantization levels.
2. **Quantization Error (Q_e):** Because a continuous voltage is rounded to the nearest discrete digital level, an inherent rounding error exists in all ADCs (Kumar, p. 957):

$$Q_e = \pm (1 / 2) \text{ LSB}$$

This error can never be eliminated by circuit tuning; it is reduced only by adding more bits. 3. **Conversion Time (t_c):** The total elapsed time from the command to start conversion (Start of Conversion, **SOC**) to the moment the valid digital word appears at the output pins (End of Conversion, **EOC**).

5.5 Counting (Ramp) ADC

Source: Kumar (Ch. 17, pp. 958–960, Figure 17.15)

[Pacing: Simple & Intuitive — The Staircase Comparator Method]

How It Works (The Physical Mechanism)

The Counter-Type ADC uses a **closed-loop feedback system** consisting of an analog comparator, a binary counter, and an internal DAC (Kumar, p. 958, Figure 17.15):

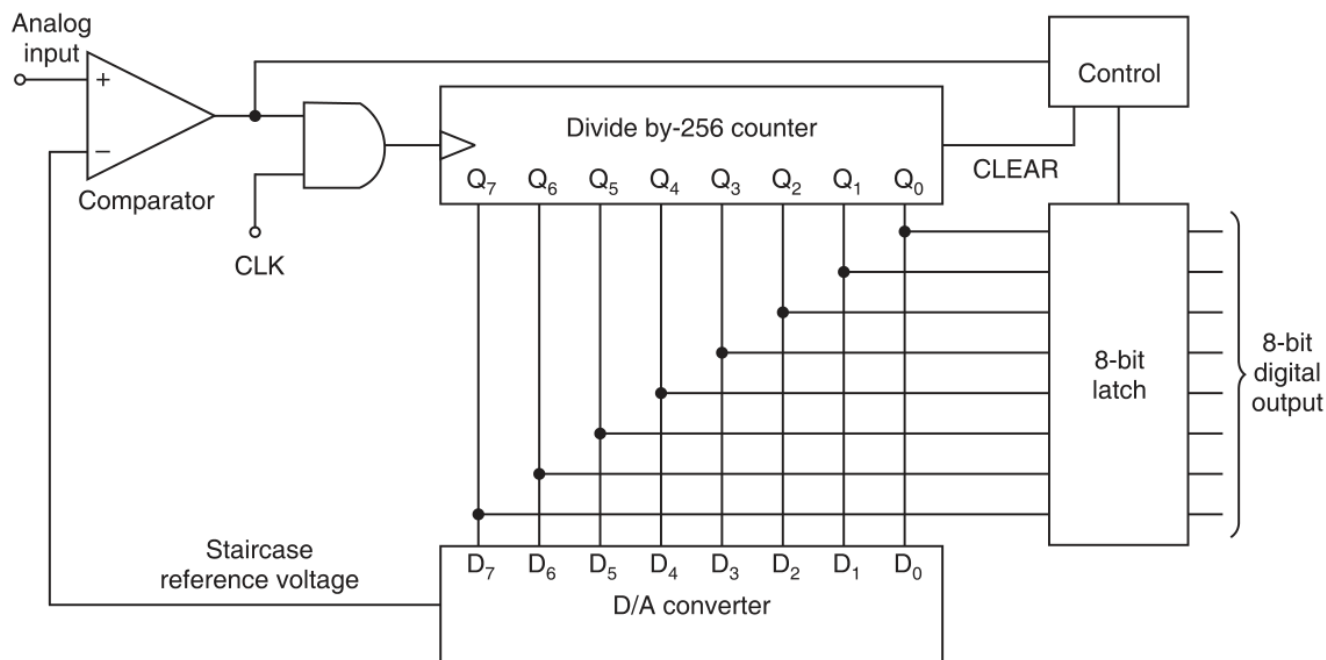


Figure 17.15 Logic diagram of the counter-type ADC.

Figure

17.15: Logic schematic of a Counter-Type (Ramp) Analog-to-Digital Converter. Sourced from A. Anand Kumar, Fundamentals of Digital Circuits (4th Ed.), p. 958 (printed p. 926).

Characteristic 13 Walkthrough: Hardware Reality Check

1. A **Start pulse** resets the binary counter to 0000. The DAC output voltage is $V_{\text{DAC}} = 0 \text{ V}$.
2. The analog input voltage V_A is connected to the non-inverting (+) input of the comparator, and V_{DAC} connects to the inverting (-) input.
3. Because $V_A > V_{\text{DAC}}$, the comparator outputs **HIGH (1)**, which enables an AND gate, allowing master clock pulses to enter the counter.
4. The counter counts upward (0, 1, 2, 3 dots), causing the internal DAC output V_{DAC} to climb upward like a **staircase ramp**.
5. The instant V_{DAC} climbs just a fraction of a millivolt above V_A ($V_{\text{DAC}} \geq V_A$), the comparator output **snaps LOW (0 V)**!
6. The AND gate is instantly inhibited, freezing the clock. The counter stops counting.
7. The binary word frozen in the counter is the exact digital equivalent of the input voltage V_A !

Limitation: Variable and Slow Conversion Time

- If V_A is small (e.g., 0.1 V), the counter stops in 5 clock cycles.
- If V_A is full-scale, an 8-bit counter must count through all $2^8 = 256$ clock cycles!
- In an n -bit counter, the maximum conversion time is:

$$t_{\text{max}} = 2^n \times T_{\text{clock}}$$

This massive variation in conversion time makes the counting ADC too slow for audio or video.

5.6 Successive Approximation ADC (SAR ADC)

Source: Kumar (Ch. 17, pp. 965–967, Figure 17.24); Floyd (Ch. 9, pp. 494–498)

[Pacing: Conceptually Heavy — The Binary Search Algorithm in Hardware]

Need (Problem-First)

We need an ADC that is fast, highly accurate, and has a **fixed, predictable conversion time** that does not depend on whether the input voltage is high or low.

The Binary Search Algorithm in Hardware (Kumar, p. 965, Figure 17.24)

Instead of counting up slowly by 1s (like a linear search), the SAR ADC uses an intelligent **binary search algorithm** (the same strategy you use when guessing a secret number between 1 and 100 by guessing 50 first):

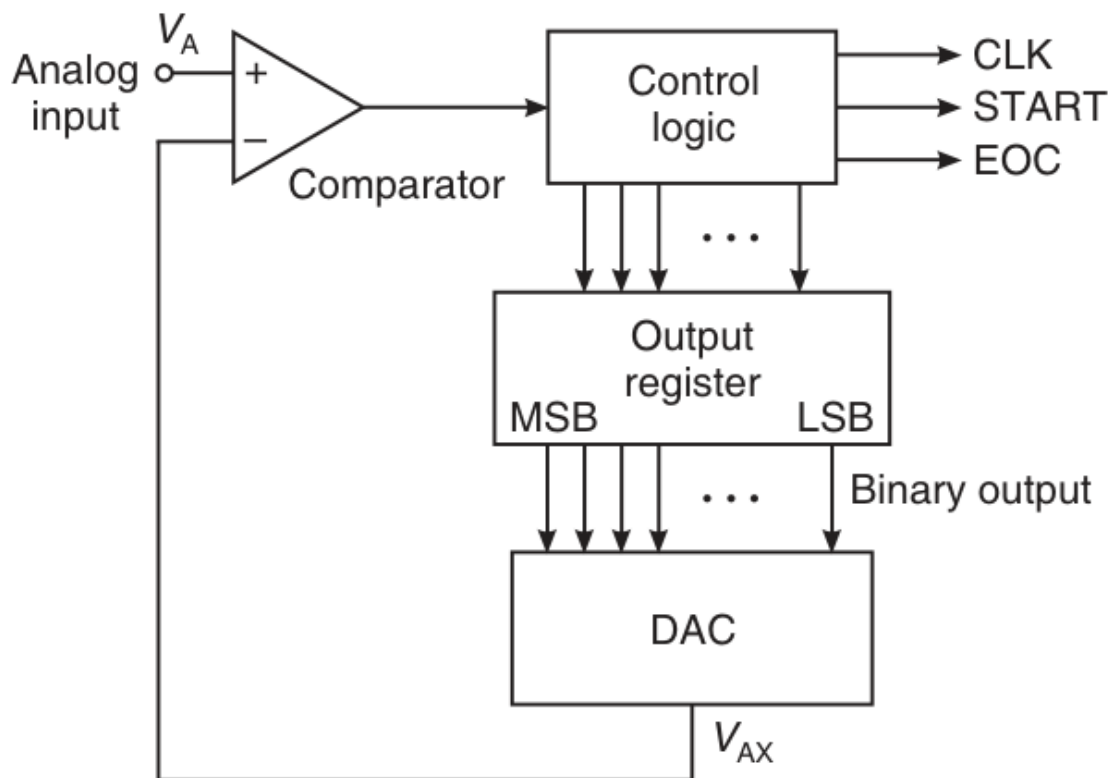


Figure 17.24 The successive-approximation type ADC.

Figure

17.24: Block diagram of a Successive-Approximation Register (SAR) Analog-to-Digital Converter. Sourced from A. Anand Kumar, Fundamentals of Digital Circuits (4th Ed.), p. 965 (printed p. 933).

Characteristic 13 Walkthrough: Step-by-Step Conversion

Consider a 4-bit SAR ADC with full-scale $V_{\text{FS}} = 16 \text{ V}$ (each step is 1 V).

Suppose the incoming analog voltage is $V_A = 11.4 \text{ V}$. - **Clock Cycle 1 (Test MSB, $D_3 = 8 \text{ V}$):**

SAR sets MSB to 1: trial word = 1000 (8 V).

Internal DAC produces 8 V .

Comparator tests: Is $V_A (11.4 \text{ V}) > 8 \text{ V}$? YES!

Decision: Keep $D_3 = 1$. - **Clock Cycle 2 (Test Next Bit, $D_2 = 4 \text{ V}$):**

SAR sets D_2 to 1: trial word = 1100 ($8 + 4 = 12 \text{ V}$).

Internal DAC produces 12 V .

Comparator tests: Is $V_A (11.4 \text{ V}) > 12 \text{ V}$? NO! (It's too high!)

Decision: Reset $D_2 = 0$. - **Clock Cycle 3 (Test Next Bit, $D_1 = 2 \text{ V}$):**

SAR sets D_1 to 1: trial word = 1010 ($8 + 0 + 2 = 10 \text{ V}$).

Internal DAC produces 10 V .

Comparator tests: Is $V_A (11.4 \text{ V}) > 10 \text{ V}$? YES!

Decision: Keep $D_1 = 1$. - **Clock Cycle 4 (Test LSB, $D_0 = 1 \text{ V}$):**

SAR sets D_0 to 1: trial word = 1011 ($8 + 0 + 2 + 1 = 11 \text{ V}$).

Internal DAC produces 11 V .

Comparator tests: Is $V_A (11.4 \text{ V}) > 11 \text{ V}$? YES!

Decision: Keep $D_0 = 1$. - **End of Conversion (EOC):** The SAR asserts the EOC pin. Final digital word = 1011 (11_{10})! -

Fixed Conversion Time: Notice that for an n -bit SAR ADC, conversion **always takes exactly n clock cycles**!

$$t_{\text{conversion}} = n \times T_{\text{clock}}$$

An 8-bit SAR ADC finishes in just **8 clock cycles**, over **30 times faster** than a counting ADC!

Commercial ADC IC: ADC0801 / ADC0808 (Kumar, p. 967, Figure 17.25)

A classic 8-bit successive approximation ADC with on-chip clock generator, active-low chip select (**CS**), write/start (**WR**), and end-of-conversion interrupt (**INTR**).

5.7 Dual-Slope Integrating ADC

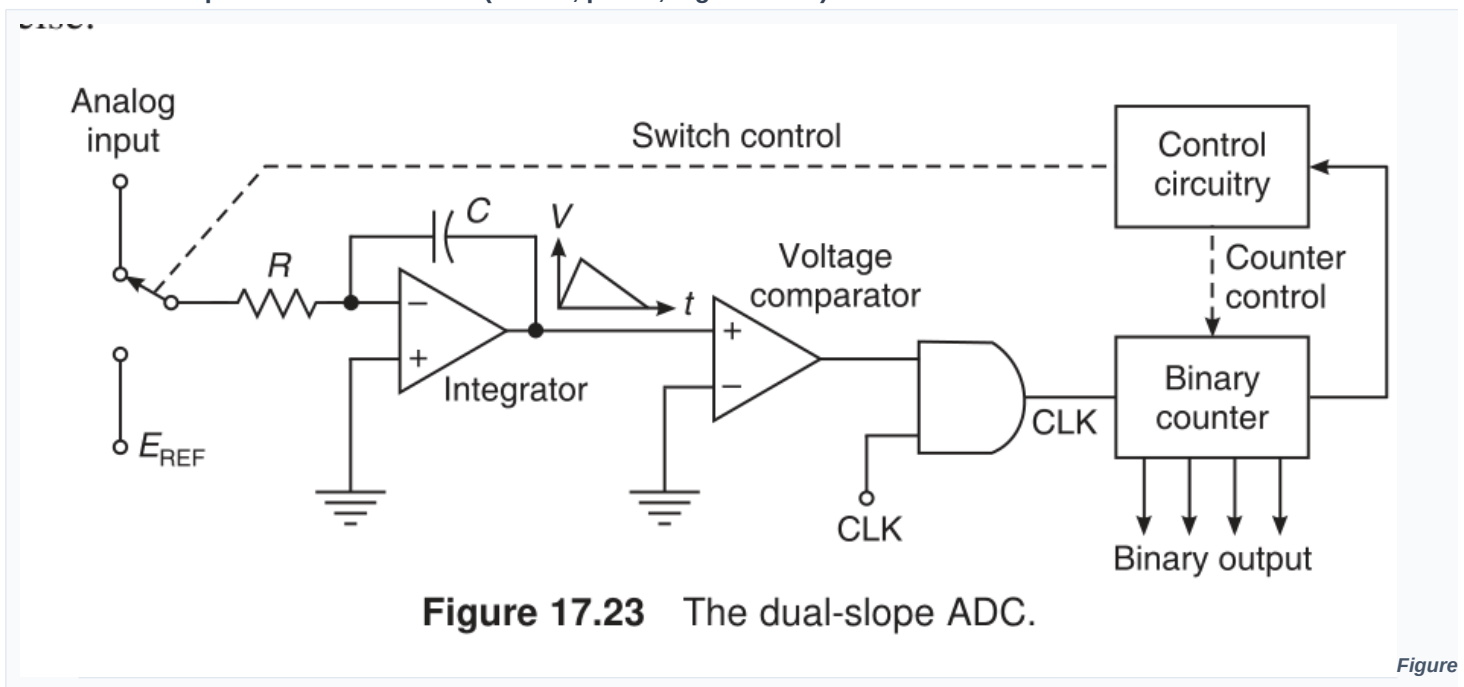
Source: Kumar (Ch. 17, pp. 964–965, Figure 17.23); Floyd (Ch. 9, pp. 498–500)

[Pacing: Mathematical Masterpiece — The Heart of Digital Multimeters]

Need (Problem-First)

In industrial and medical measurement instruments (like digital multimeters), high speed is not required, but **extreme precision and total immunity to 50 Hz / 60 Hz AC power line noise** are mandatory. SAR ADCs suffer from component drift in their DAC resistors. We need an ADC whose mathematical accuracy is **completely independent of component tolerances** (independent of **R** and **C**)! That converter is the **Dual-Slope Integrating ADC**.

The Dual-Slope Circuit & Waveform (Kumar, p. 964, Figure 17.23)



17.23: Functional block diagram of the Dual-Slope Integrating ADC. Sourced from A. Anand Kumar, Fundamentals of Digital Circuits (4th Ed.), p. 964 (printed p. 932).

The Two Mathematical Integration Phases:

1. Phase 1: Run-Up (Fixed Time Interval T_1):

The electronic switch connects the op-amp integrator to the unknown analog input voltage V_{in} (assumed negative, or inverted). The integrator charges its capacitor **C** through resistor **R** for a **fixed time period T_1** (governed by the time it takes an **n**-bit counter to count to full scale, 2^n clock cycles):

$$V_{\text{peak}} = -(1 / RC) \int_0^{T_1} V_{\text{in}} dt = \frac{V_{\text{in}} \cdot T_1}{RC}$$

Notice that the peak voltage reached is directly proportional to V_{in} . 2. **Phase 2: Run-Down (Variable Time Interval T_2):**

At the exact moment T_1 expires, the control logic flips the switch to a fixed, highly stable reference voltage $-V_{\text{ref}}$ of opposite polarity. The counter resets to zero and starts counting again. The integrator discharges back toward zero:

$$0 = V_{\text{peak}} - \frac{V_{\text{ref}} \cdot T_2}{RC}$$

Equating the two expressions:

$$\frac{V_{in} \cdot T_1}{RC} = \frac{V_{ref} \cdot T_2}{RC}$$

Divide both sides by RC :

$$V_{in} \cdot T_1 = V_{ref} \cdot T_2 \text{ iff } \mathbf{V_{in} = V_{ref} \leq ft((T_2 / T_1)right)}$$

The Mathematical Miracle of Dual-Slope:

Look at that equation! The resistance R , capacitance C , and master clock frequency f have completely cancelled out of the equation!

Even if your resistor shifts by **10%** due to heat or your capacitor ages, the measured voltage V_{in} remains **100% mathematically exact!** - By setting T_1 to an exact multiple of the AC power line period (**20 ms** for **50 Hz** or **16.67 ms** for **60 Hz**), AC line hum integrates to exactly zero, providing infinite noise rejection.

PART 3: Semiconductor Memories and Programmable Logic Devices (PLDs)

5.8 Memory Organization, Word Capacity, and Read/Write Operations

Source: Kumar (Ch. 18, pp. 974–978, Figures 18.1 & 18.2); Mano (Ch. 7, pp. 410–418)

[Pacing: Architectural System View — High Relevance to Computer Hardware]

Memory Terminology Defined Inline:

- **Memory Cell:** The elementary 1-bit bistable storage unit.
- **Word:** A group of binary bits that are stored or retrieved together as a unit (e.g., 8-bit byte, 16-bit word, 32-bit word, 64-bit word).
- **Address:** The unique binary number designating the specific physical physical location of a word inside the memory array.
- **Capacity:** The total storage capability of the memory, designated as:

$$\text{Capacity} = 2^k \times m$$

Where k is the number of **Address lines** (giving 2^k addressable words) and m is the number of **Data lines** (bits per word) (Kumar, p. 975). Example: A memory chip with 10 address lines and 8 data lines has:

$$\text{Capacity} = 2^{\{10\}} \times 8 = 1024 \times 8 = 1 \text{ Kilobyte (1 KB)}$$

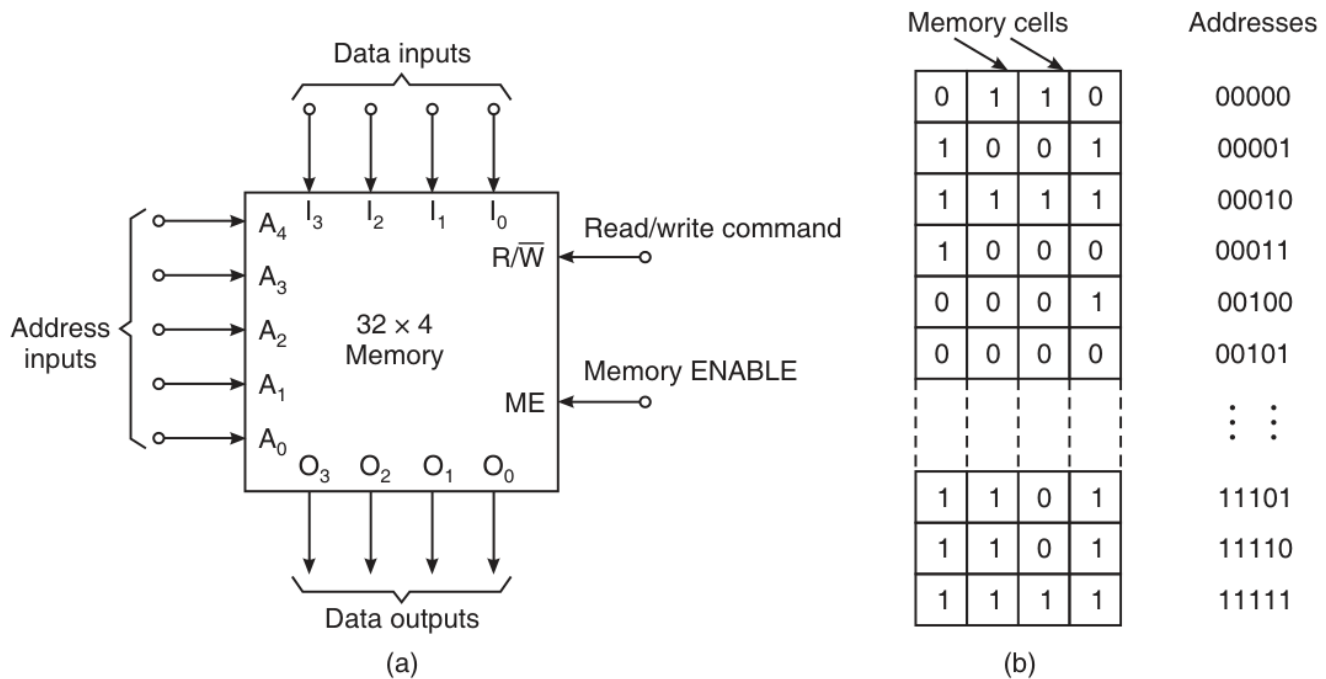


Figure 18.1 Diagram of a 32×4 memory and arrangement of memory cells.

Figure

18.1: Block diagram of a 32×4 semiconductor memory array showing internal address decoding and cell matrix. Sourced from A. Anand Kumar, *Fundamentals of Digital Circuits (4th Ed.)*, p. 976 (printed p. 944).

The Memory Read and Write Sequences (Kumar, p. 977):

- **Memory Write Cycle:**
- Apply the binary address to the address bus pins (A_0 to A_{k-1}).
- Place the data bits to be stored onto the data input bus (D_0 to D_{m-1}).
- Assert the active-low Chip Select (**CS**) pin to enable the chip.
- Pulse the Write Enable (**WE**) line LOW for a specified minimum write pulse width (t_w). The internal gates latch the data into the addressed cells.
- **Memory Read Cycle:**
- Apply the desired binary address to the address bus pins.
- Assert Chip Select (**CS**) and Output Enable (**OE**).
- Hold Write Enable (**WE**) HIGH (Read mode).
- After an **Access Time** (t_{acc}), the data stored in the addressed cells appears on the output pins.

5.9 Expanding Memory Size: Word Length and Word Capacity

Source: Kumar (Ch. 18, pp. 984–989, Figures 18.7 & 18.9); Floyd (Ch. 9, pp. 505–510)

[Pacing: Systematic Circuit Synthesis — Learn How Chips are Combined]

Need (Problem-First)

In an embedded system, your processor requires a memory space of $1K \times 8$ (1024 bytes), but the only chips available in your inventory are $1K \times 4$ (each storing 4-bit nibbles). Or your system needs $2K \times 8$, but you only have $1K \times 8$ chips. How do you wire multiple small chips together to expand the total memory space?

1. Expanding Word Length (Increasing Data Width) (Kumar, p. 985)

- **Goal:** Combine two 16×4 memory chips to create a single 16×8 memory module.
- **The Wiring Rule:**
- Both chips must share the **exact same address bus lines** (A_0 , A_1 , A_2 , A_3) in parallel so that identical memory locations are accessed simultaneously!
- Both chips share the **exact same control lines** (**CS**, **WE**).

- Chip 1 provides the lower 4 data bits (**D_0, D_1, D_2, D_3**).
- Chip 2 provides the upper 4 data bits (**D_4, D_5, D_6, D_7**).

for each 16K of 8-bit words or a total of $2 \times (128K/16K) = 16$ circuits.

Figure 18.7 shows how two 16×4 memory RAM chips with common I/O lines are combined to produce a 16×8 memory. Since each chip can store 16 4-bit words and since 16 8-bit words are to be stored, each chip is used to store half of each word. In other words, RAM 0 stores the four higher order bits of each of the 16 words, and RAM 1 stores the four lower order bits of each of the 16 words. A full 8-bit word is available at the RAM outputs connected to the data bus.

Figure

18.7: Combining two 16×4 RAM chips to expand word length, creating a unified 16×8 memory module. Sourced from A. Anand Kumar, *Fundamentals of Digital Circuits (4th Ed.)*, p. 985 (printed p. 953).

2. Expanding Word Capacity (Increasing Total Addresses) (Kumar, p. 988)

- **Goal:** Combine two 16×4 memory chips to create a single 32×4 memory module.
- **The Wiring Rule:**
 - The module needs $32 = 2^5$ words ■ **5 address lines** (**A_4, A_3, A_2, A_1, A_0**).
 - The lower 4 address lines (**A_0, A_1, A_2, A_3**) are connected to both chips in parallel.
 - The data lines (**D_0, D_1, D_2, D_3**) of both chips are tied together in parallel to form a common data bus.
 - The highest address bit (**A_4**) is used to select which chip is active!
 - When **A_4 = 0**: Chip 1 is enabled via **CS_1** (addresses **0** to **15**).
 - When **A_4 = 1**: Chip 2 is enabled via **CS_2** through an inverter (addresses **16** to **31**).

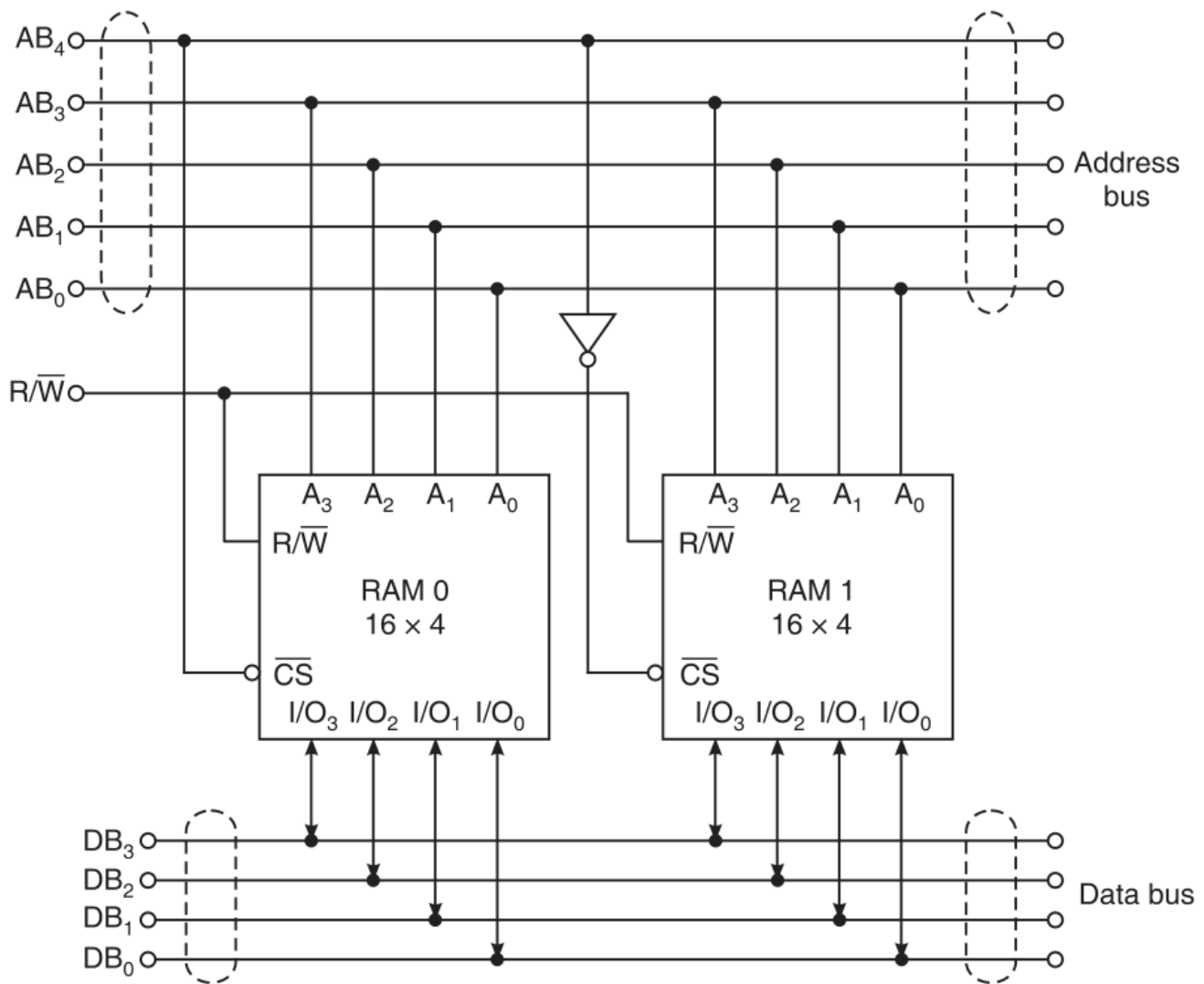


Figure 18.9 Combining two 16×4 chips for a 32×4 memory.

Figure

18.9: Combining two 16×4 RAM chips using an MSB address inverter to expand word capacity to 32×4 . Sourced from A. Anand Kumar, *Fundamentals of Digital Circuits (4th Ed.)*, p. 988 (printed p. 956).

5.10 Classification and Characteristics of Memories: SRAM vs. DRAM, ROM Families

Source: Kumar (Ch. 18, pp. 978–984, Figure 18.5); Floyd (Ch. 9, pp. 510–525)

[Pacing: Core Memory Technologies — Understand the Silicon Cells]

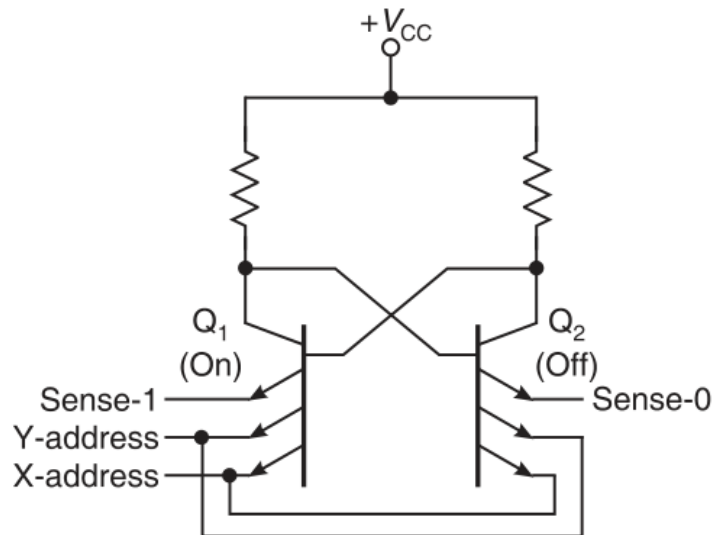
The Master Memory Hierarchy:

- **Volatile Memory:** Loses its stored data the instant electric power is disconnected. (RAM).
- **Non-Volatile Memory:** Retains its stored binary data permanently, even with zero power applied. (ROM, Flash).

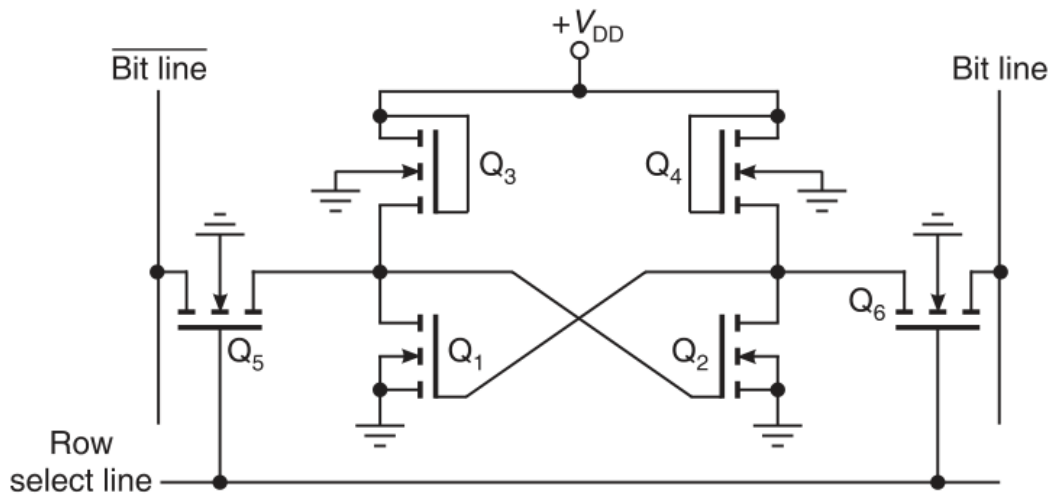
1. Static RAM (SRAM) vs. Dynamic RAM (DRAM)

Parameter	Static RAM (SRAM)	Dynamic RAM (DRAM)
Storage Element	Cross-coupled flip-flop (4 to 6 transistors)	Single MOSFET transistor + tiny capacitor (1T-1C)
Silicon Cell Area	Large (takes 4x more area per bit)	Microscopic (ultra-dense)

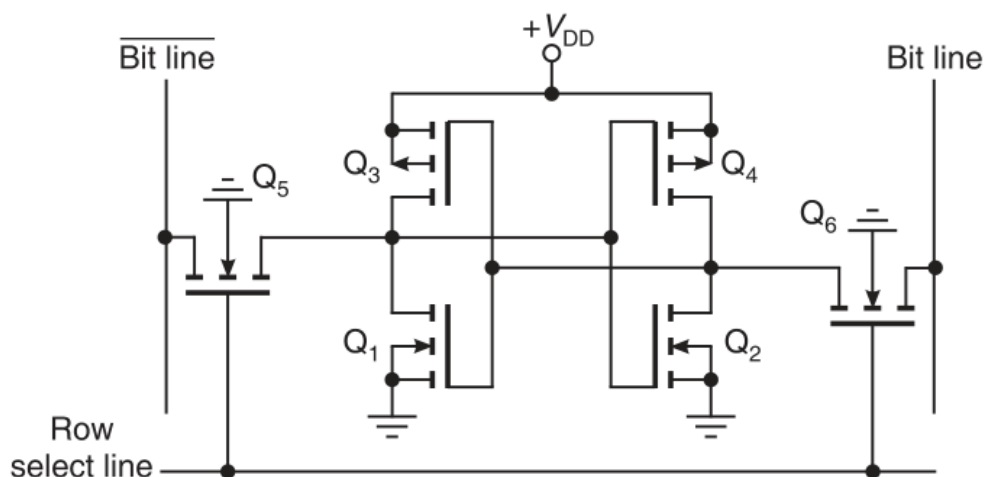
Parameter	Static RAM (SRAM)	Dynamic RAM (DRAM)
Refresh Requirement	None (stores data indefinitely while powered)	Mandatory periodic refresh (every few ms)
Operating Speed	Extremely fast (1 to 5 ns)	Slower (10 to 50 ns)
Power Consumption	Higher when switching; zero refresh overhead	Lower per bit; requires refresh controller power
Application	CPU L1/L2/L3 Caches	Main System Memory (Computer RAM sticks)



(a) A bipolar (TTL) RAM cell constructed from multi-emitter transistors cross connected to form a latch



(b) NMOS RAM cell



(c) CMOS RAM cell

Figure 18.5 The bipolar, NMOS, and CMOS RAM cells.

Figure 18.5: Circuit

Why DRAM Needs Refreshing:

In DRAM, a bit 1 is stored as a tiny packet of electrostatic charge on an integrated capacitor (≈ 30 femtofarads). Due to microscopic reverse-bias junction leakage, this charge bleeds away into the silicon substrate within tens of milliseconds! If left alone, a stored 1 will bleed down into a 0. Therefore, a dedicated **DRAM Controller** must continuously read every row and restore the full charge thousands of times per second (the **Refresh Cycle**).

2. The Read-Only Memory (ROM) Hierarchy

- **Mask ROM:** Data is permanently programmed into the silicon metallization layer during factory fabrication with chemical masks. Cheap in million-unit quantities, but cannot be modified.
 - **PROM (Programmable ROM):** Manufactured with microscopic fusible links (nichrome or polysilicon). The user programs data in a PROM programmer by pulsing high current to physically **blow open selected fuses** ("burning" the ROM). It can be programmed only once!
 - **EPROM (Erasable PROM):** Uses floating-gate MOS transistors. It can be erased by exposing the quartz window on the chip package to intense ultraviolet (UV) light for 20 minutes, which discharges the floating gates.
 - **EEPROM (Electrically Erasable PROM):** Can be erased and rewritten electrically byte-by-byte in circuit, without UV light.
 - **Flash Memory:** A high-density evolution of EEPROM that erases and writes data in whole blocks (pages), powering USB thumb drives and modern Solid-State Drives (SSDs).
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5.11 Programmable Logic Devices (PLDs): PROM, PAL, and PLA

Source: Kumar (Ch. 8, pp. 491–508, Figures 8.7, 8.8, 8.15); Mano (Ch. 7, pp. 430–445)

[Pacing: Conceptually Heavy — The Three Architectures of Reconfigurable Silicon]

Need (Problem-First)

In the 1970s, a complex digital product (like an arcade game or cash register) required hundreds of discrete 7400-series TTL chips wired together on massive printed circuit boards. Every design change required redesigning and etching a new circuit board. Engineers wanted a single standardized blank chip that could be configured on a programmer to implement *any arbitrary collection of logic functions*. These chips are **Programmable Logic Devices (PLDs)**.

The Three Core Architectures Compared (Kumar, p. 499, Figure 8.7)

Every two-level Boolean function can be expressed in Sum-of-Products (SOP) form:

$$F = \sum (\text{Product terms}) = \text{OR of ANDs}$$

Therefore, a PLD consists of an array of **AND gates** (generating product terms) followed by an array of **OR gates** (summing the product terms).

The difference between the three PLD types lies strictly in **which array is fixed and which array is programmable**:

y summed in each OR gate. The programmable array logic (PAL) has programmable

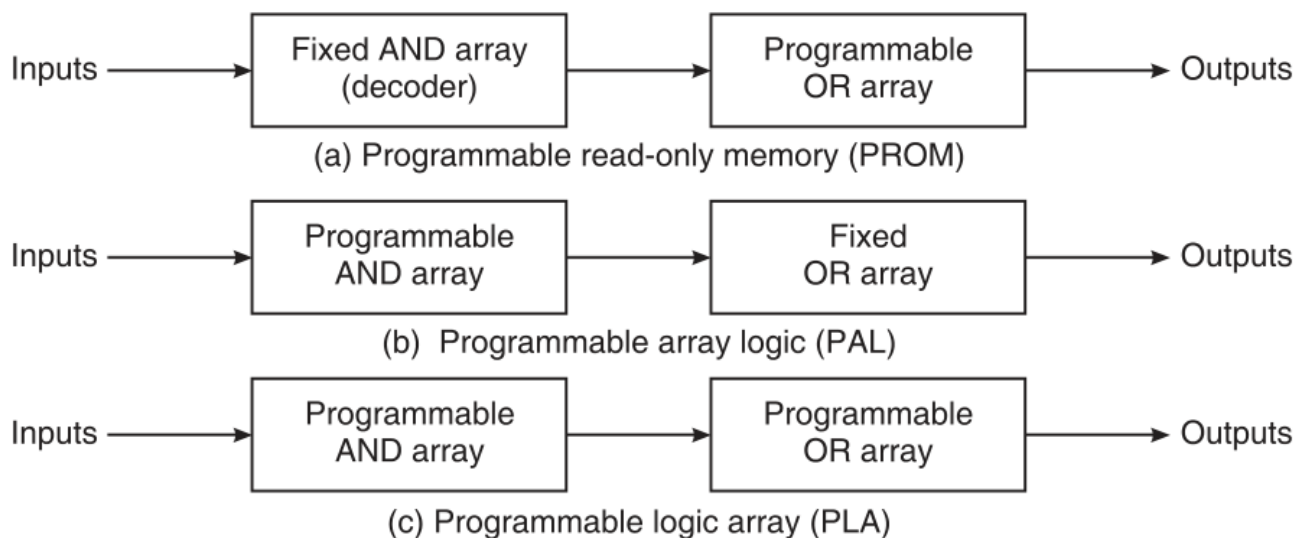


Figure 8.7 Basic configuration of three PLDs.

Figure

8.7: Architectural comparison of the three fundamental Programmable Logic Devices: PROM, PAL, and PLA. Sourced from A. Anand Kumar, *Fundamentals of Digital Circuits (4th Ed.)*, p. 499 (printed p. 467).

PLD Type	AND Array	OR Array	Flexibility	Ease of Programming
PROM	Fixed (Full Decoder)	Programmable	Low	Very Easy
PAL	Programmable	Fixed	Moderate	High (Industry Favorite)
PLA	Programmable	Programmable	Maximum	More Complex

1. Programmable Array Logic (PAL) (Kumar, p. 500, Figure 8.8)

Invented by Monolithic Memories Inc. (MMI).

- **Architecture:** The AND array is fully programmable (inputs and complements can be fused to any AND gate). The OR array is **hardwired and fixed** (each OR gate receives product terms from a fixed group of AND gates, e.g., 3 or 4 AND gates per OR gate):

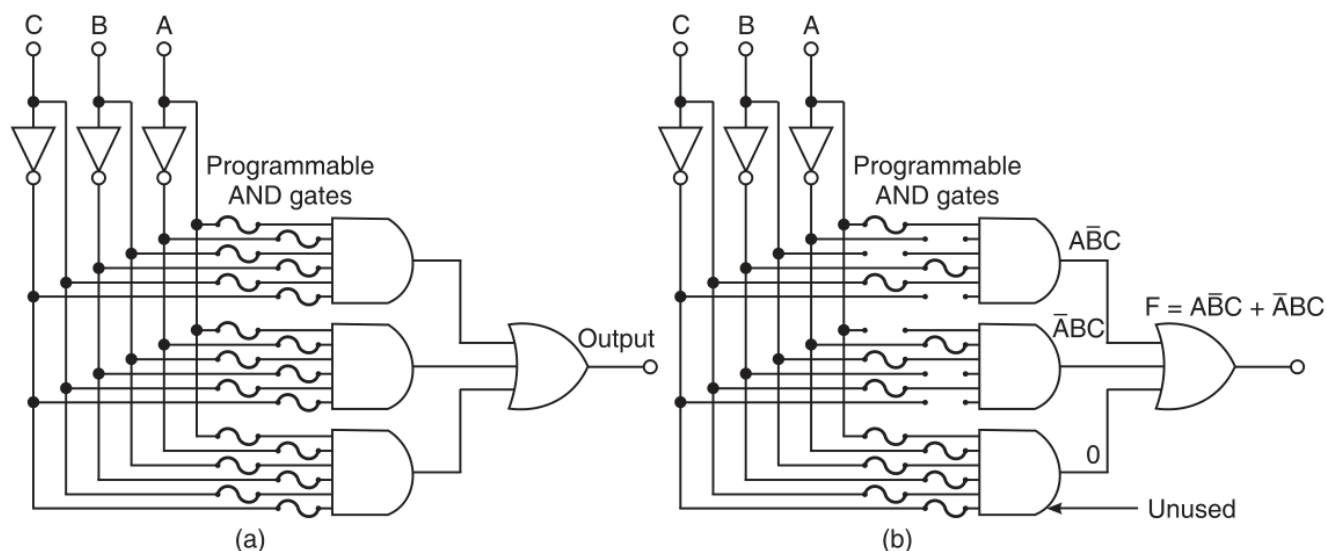


Figure 8.8 Basic structure of a PAL circuit, and implementation of $F = A\bar{B}\bar{C} + \bar{A}BC$.

Figure

8.8: Basic structure of a Programmable Array Logic (PAL) circuit showing unprogrammed fusible links (a) and programmed implementation of $F = A\bar{B}\bar{C} + \bar{A}BC$ (b). Sourced from A. Anand Kumar, *Fundamentals of Digital Circuits (4th Ed.)*, p. 500 (printed p. 468).

- **Advantage:** Eliminating the programmable OR array drastically reduces parasitic propagation delay, making PALs exceptionally fast and simple to program.

2. Programmable Logic Array (PLA) (Kumar, p. 508, Figure 8.15)

- **Architecture:** Both the AND array and the OR array are fully programmable:

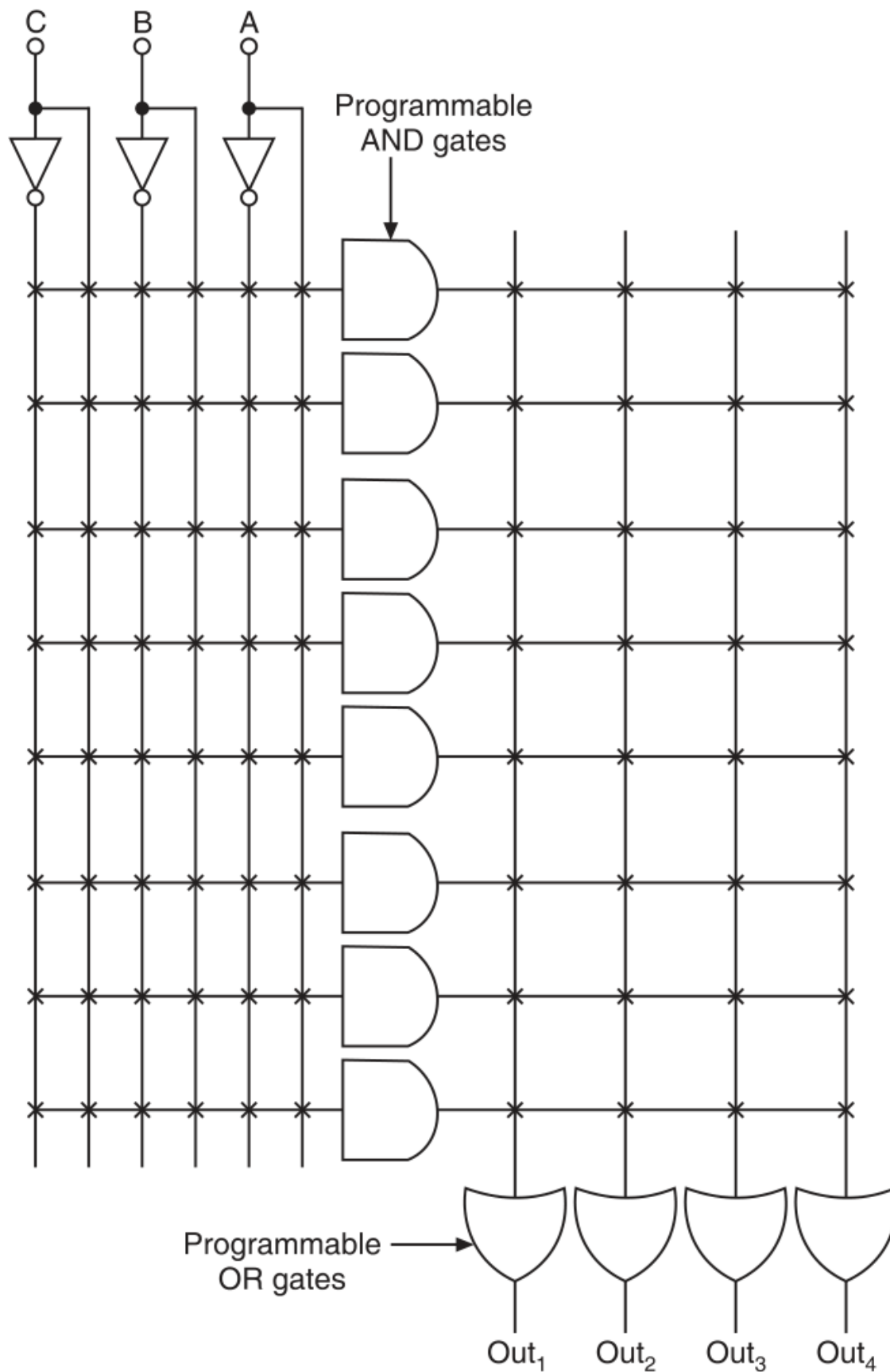


Figure 8.15 Structure of (an unprogrammed) PLA circuit.

Figure 8.15:

Internal matrix layout of an unprogrammed Programmable Logic Array (PLA) circuit showing programmable AND crosspoints and programmable OR crosspoints. Sourced from A. Anand Kumar, *Fundamentals of Digital Circuits* (4th Ed.), p. 508 (printed p. 476).

- **Why PLA is Superior for Complex Shared Systems:**

Because the OR array is programmable, multiple outputs can **share the exact same product term** from the AND array! If two functions both require the term **ABC**, a PLA generates **ABC** once in the AND array and routes it to both OR gates, saving silicon area.

Unit V Understanding Checkpoints & Misconception Audit

1. Does This Make Sense? Self-Check

- **Question 1:** Why is an R-2R ladder preferred over a binary weighted-resistor network for high-resolution DACs?
- **Answer:** Because an R-2R ladder requires only two precision resistor values (**R** and **2R**), avoiding the unmanageable resistor spread (over **2000:1** in 12-bit DACs) of weighted-resistor networks.
- **Question 2:** Why is the conversion accuracy of a Dual-Slope ADC completely independent of its resistor **R** and capacitor **C** values?
- **Answer:** Because both the charging run-up phase (**T₁**) and discharging run-down phase (**T₂**) occur through the exact same **RC** network, causing **R** and **C** to mathematically cancel out completely from the final voltage transfer equation.
- **Question 3:** What is the architectural difference between a PAL and a PLA?
- **Answer:** A PAL has a programmable AND array with a fixed OR array. A PLA has both a programmable AND array and a programmable OR array.

2. Conceptual Misconceptions Corrected

- **Misconception:** "Dynamic RAM (DRAM) is faster than Static RAM (SRAM) because it is called 'dynamic'."
Correction: The exact opposite is true! SRAM is significantly faster (**1 ext{ to }5 ext{ ns}**) because it uses active flip-flop latches. DRAM is slower (**10 ext{ to }50 ext{ ns}**) because charging and reading tiny capacitors takes time, and periodic refresh cycles block memory access.
 - **Misconception:** "A 16-bit ADC has zero error because 16 bits is very high precision."
Correction: Every ADC has an unavoidable fundamental **Quantization Error of $\pm \frac{1}{2} \text{ ext{ LSB}}$** caused by rounding continuous voltages to discrete binary levels.
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